DTAF SUPPORTING MATERIALS

This document provides more in-depth information on the individual frameworks that compose the DTAF methodology.

1. DTC Capability Periodic Table

Scope and functionality are critical when designing and implementing Digital Twins (DTs). A DT must be constructed not as an isolated model or visualization, but as an intelligent, data-driven system that evolves with its physical counterpart and supports real-world decision-making. To achieve this, organizations require structured guidance on defining what a DT should do, how intelligent or autonomous it should be, and which data or processes are necessary to enable that functionality. The Digital Twin Consortium (DTC) provides such guidance through its combined Capabilities Periodic Table (CPT) [1] and AI Agent Capabilities Periodic Table (AIA CPT) [2], two complementary frameworks that together form a comprehensive capability-based design and assessment methodology.

The DTC Capabilities Framework establishes a technology-agnostic, use-case-oriented approach for building DTs and their supporting intelligent agents. Rather than focusing on specific software tools, sensors, or vendors, it emphasizes the capabilities required to achieve a defined business or operational outcome. This shift from tool-centric to capability-centric thinking enables organizations to systematically define the scope and functionality of DT, determining which functions are essential, which are operational, and how these functions evolve as DT matures. Using the periodic table format, the DTC framework maps the full range of DT capabilities across domains, offering a structured vocabulary for users to collaboratively design and benchmark their digital ecosystems.

At its core, the CPT promotes a capability-driven design approach, in which each use case defines a specific operational or decision-support objective. These use cases are translated into explicit capability requirements using the CPT, which can then be aggregated to determine broader system needs, platform requirements, and technology solutions. This ensures that DT development remains grounded in functional purpose and measurable outcomes, rather than a fragmented, tool-driven implementation.

Building on this foundation, the AIA CPT extends the CPT framework by introducing AI-enabled interactivity to assist in the generation, customization, and assessment of capability models. While the CPT provides a static map of possible DT capabilities, the AIA CPT transforms it into an intelligent interactive toolkit that helps organizations rapidly tailor capability frameworks to their specific use cases. By inputting project details or operational objectives, users can receive AI-generated recommendations for relevant CPT capabilities, prioritized rankings, maturity-level assessments, and implementation roadmaps.

Together, the CPT and AIA CPT form a unified, use-case-oriented framework that bridges the gap between physical systems, digital representations, and cognitive automation. The CPT provides the base map of required capabilities, while the AIA CPT functions as the AI-powered companion that helps generate and operationalize those capabilities within specific organizational contexts. This combination ensures that DT projects are not only well-scoped and technically grounded but also adaptable, scalable, and strategically aligned. By guiding teams from high-

level business objectives to detailed capability specifications, the DTC Capability Framework enables systematic, efficient, and trustworthy DT implementation across sectors and maturity levels. Figure 1 illustrates the DTC periodic table.

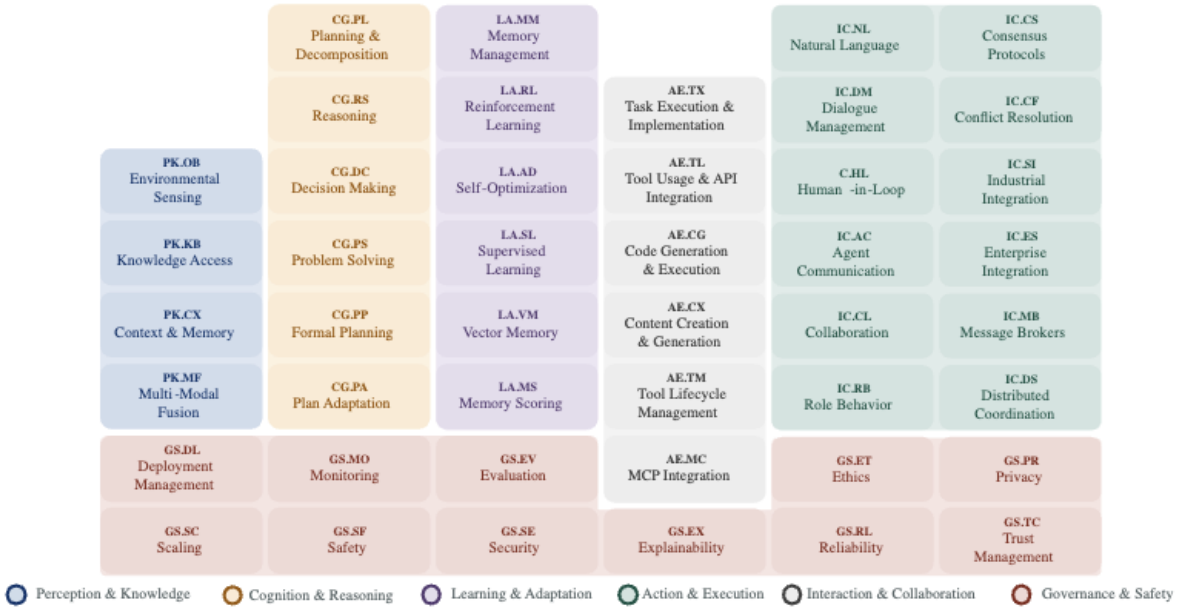


Figure 1: DTC Periodic Table of Digital Twin Capabilities.

2. DEMA

Data readiness is foundational for any DT, as unmapped or hidden data flows can compromise its accuracy and utility. Studies have shown that more than 90% of organizational data flows are unknown at the system level, often hidden within siloed knowledge. Therefore, any framework for DT or Digital Thread implementation should account for hidden data and information flows to accurately reflect real-world systems. To address this, Data Element Mapping and Analysis (DEMA) provides a systematic approach for identifying, capturing, and mapping data flows in complex systems [3,4]. DEMA helps organizations understand where their data truly originates and where it ultimately ends up.

Unlike traditional methods that focus on documents or high-level processes, DEMA dives deeper by examining data at the data element level, the smallest meaningful units, such as part numbers, approval dates, or design dimensions. By doing this, DEMA uncovers hidden or manual data exchanges that often disrupt system efficiency and reliability. It creates transparency by showing how individual data elements connect across departments, software platforms, and lifecycle stages, enabling a more complete and accurate digital thread.

The DEMA process follows three steps. The first step, the Functional-Level Mapping, identifies the major functional areas and sub-activities that make up an organization's workflow. This step establishes the foundation for understanding where data are created, used, and exchanged. Functional-level maps show departments interact and how processes align with organizational goals, helping the analyst trace every piece of data to its functional purpose. The second step, Data Vessel-Level Mapping, dives deeper by documenting the movement of containers, called data vessels, across these functions. Data vessels include spreadsheets, reports, CAD files, emails, or even handwritten notes. It can be anything that stores or transfers

data. In this step, DEMA identifies how these vessels are shared, where they are stored, and who has access to them. By capturing these flows, organizations can see how data travels between people and systems and where efficiencies of “dead ends” exist. The third step, Data Element-Level Mapping, is where each data vessel identified in the second step is broken down into its smallest components, individual data elements. These elements might include measurements, serial numbers, inspection results, or timestamps. Mapping these elements allows the analyst to trace exactly how data is created, modified, and reused across systems and departments.

By following these three steps, DEMA transforms hidden, fragmented data into a transparent, traceable network that supports digital engineering initiatives such as DTs. Unlike older mapping tools that focus on functions or documents, DEMA provides a data-centric approach that supports both process improvements and digital transformation. It bridges the gap between systems engineering and lean manufacturing by uncovering inefficiencies, reducing redundant work, and supporting continuous improvement. Figure 2 illustrates the DEMA framework.

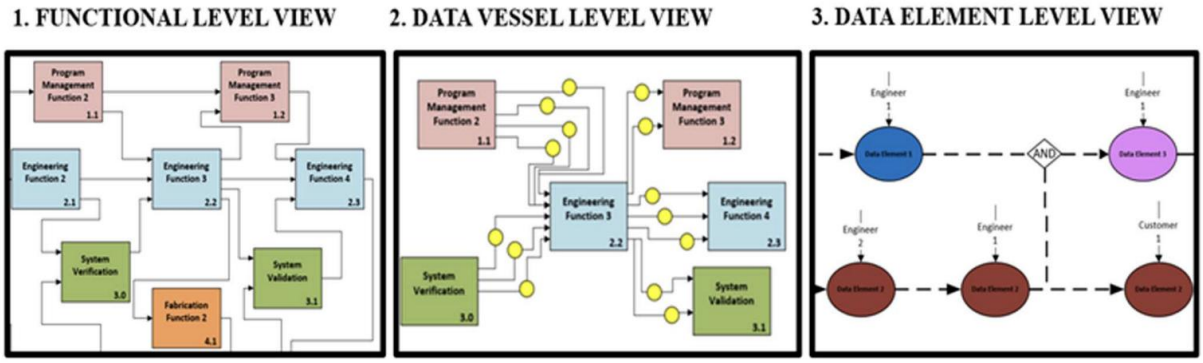


Figure 2: DEMA Overview.

3. 4R Framework

DT maturity assessment ensures that the DT is developed to the appropriate level of capability for the intended use case. Since not every use case requires a highly complex DT, organizations must identify the required level of DT capability for their technical requirements, pertinent data, and operational goals. Establishing clear objectives for creating the DT provides clear direction to ensure that the desired performance is achieved. To address this, research was performed to characterize the development of a DT based on its capabilities and maturity as a means of creating general-purpose and modular DTs, called the 4R framework [5,6]. This framework was designed to create a conceptual model that defines and guides the development of the DTs through distinct phases.

The 4R framework categorizes DTs into four phases: Representation (R1), Replication (R2), Reality (R3), and Relational (R4), each representing increasing levels of maturity and capability. The R1 phase involves understanding the physical system's behavior and setting up a data architecture and storage system. The goal is to define the features, behaviors, components, and relationships relevant for the use case, and identify what to include in the model and what not to. In the R2 phase, the system is replicated in a virtual environment using the data

architecture established in the representation phase, reproducing the same outputs given the same inputs from the physical system. It is necessary that this phase provides accurate mapping of inputs, as it is fed data from R1 and ultimately becomes the replica used to model the physical system. The R3 phase involves using the DT to explore what-if scenarios, with the aim of implementing the results obtained from virtual runs into the physical system. This creates a virtual modeling tool that can now run independently of the physical system with the ability to make predictions, provide alternative scenarios, and optimize. The R4 phase involves incorporating decision-making technologies such as AI or ML and enabling bi-directional data exchange. This allows the model to translate updates, corrections, and decisions back into the physical system, making the twin self-aware.

Each level of the 4R Framework builds upon the previous one, progressively increasing the model's analytical depth and decision-making capability. As capability advances from R1 to R4, the DT moves from the ground level of data gathering and component identification to full autonomy. This hierarchical structure demonstrates the progression from analyzing and applying data to build a digital replica, training that replica to optimize the physical system, and ultimately enabling the DT to make independent updates and corrections to the physical system, which form the foundation requirements of DT implementation. By clearly defining these stages, the 4R Framework enables researchers and practitioners to assess their desired DT capabilities, identify existing gaps, and determine the necessary level of support needed for their engineering objectives. Ultimately, the 4R framework establishes a systematic approach for transitioning from data architecture development to the creation of an intelligent, data-driven DT capable of continuous feedback, optimization, and real-time decision support. Figure 3 illustrates this ordered progression and highlights how increasing complexity correlates with the enhanced ability of the DT.

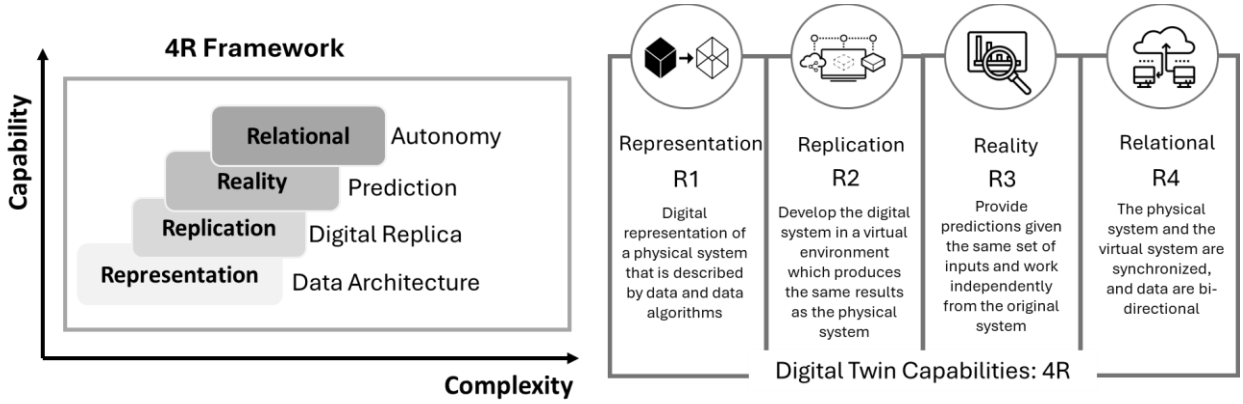


Figure 3: 4R Framework.

4. 4S Framework

Simulation capability is another critical aspect. Correctly assessing simulation levels prevents mislabeling simulations as DTs. Traditional simulation and DT models share the same capability to replicate physical systems in virtual environments, but they are not the same. Simulation is at the core of DTs, and their combined use is very common, creating confusion: many organizations mistakenly label simulation models as DTs and vice versa. While simulations have their own valuable applications, it's inaccurate to classify them as DTs. Unlike simulations,

DTs feature continuous bilateral communication, sensor-based monitoring of physical system changes, and real-time decision-making support. Research has found that a significant portion of literature claiming to combine simulation and DT fails to achieve true DT capabilities, instead presenting simulations as DTs. Even among correctly built DTs, many only utilize a fraction of their potential. When constructing a DT, it is important to identify the simulation's capabilities and ensure they fit the specific scope for each problem. Current literature has addressed this by categorizing simulation into four levels of capability and utility into one framework - The 4S Framework [7,8].

The 4S framework comprises 4 levels: Modeling (S1), Analyzing (S2), Predicting (S3), and Prescribing (S4). The S1 level involves creating an abstracted representation of a real-world system through conceptual modeling and computer modeling. Proper data collection, parameter selection, and understanding are crucial in this stage. In the S2 level, the data gathered during Modeling is incorporated into the simulation model. Input analysis ensures that the data used approximates real-world behavior. At this level, analysis is conducted to infer the system's behavior. The S3 level is where the simulation can predict system behavior under various conditions. It allows for the investigation of what-if scenarios, performance measurement, and comparison. The S4 level involves using the simulation model for decision support and optimization. Decision-makers can identify optimal courses of action, and simulation optimization or machine learning methods can be applied to enhance decision-making and system performance.

Each level of the 4S Framework builds on the previous one, progressively increasing the model's analytical depth and decision-making value. As capability advances from S1 to S4, the simulation evolves from a descriptive to a prescriptive system. This hierarchical structure not only illustrates the growth of analytical complexity but also aligns simulation maturity with the foundational requirements for DT implementation. By clearly defining these stages, the 4S Framework enables researchers and practitioners to assess their current simulation readiness, identify capability gaps, and determine the appropriate level of sophistication needed to support specific organizational or engineering objectives. Ultimately, the 4S framework provides a structured pathway for transitioning from isolated simulation models to intelligent, data-driven systems capable of continuous feedback, optimization, and real-time decision support. Figure 4 illustrates the simulation levels of capabilities and complexity.

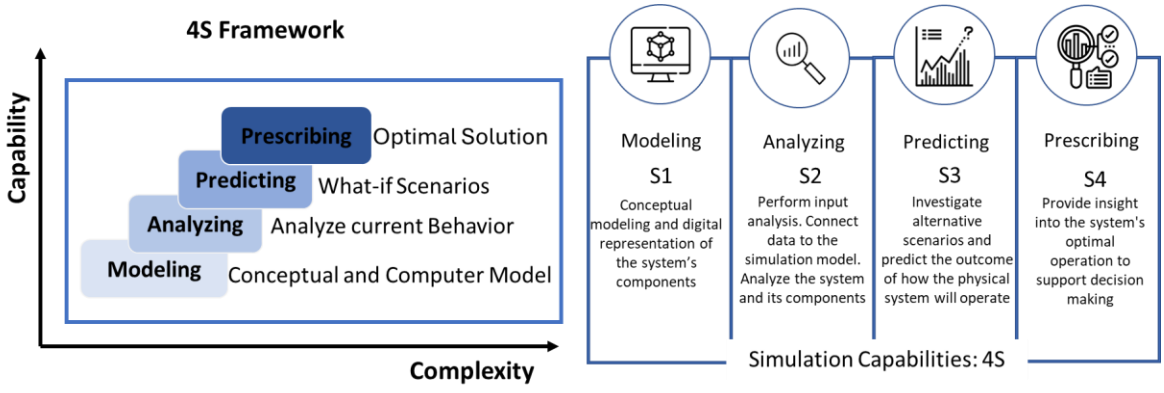


Figure 4: 4S Framework.

5. Digital Twin V

Finally, trust and credibility are essential for a DT to support real-world decision-making. It is fundamental to trust the data, accuracy, and updating procedure of the models that support decision-making, especially in manufacturing systems where cost and safety are critical. Therefore, Verification and Validation (V&V) are crucial pillars of creating trust and credibility in DTs. In this context, establishing trust includes demonstrating how the V&V evidence can be applied to support the model's use in a defined context of use. From an investment perspective, building trust in the DT implementation minimizes the risk associated with technology adoption and acceptance. One existing framework that addresses this gap is the Digital Twin V (DTV) [9, 10]. The DTV is a methodology inspired by the Vee model introduced by Forsberg and Mooz (1991), and it describes in a structured approach how to create trusted DTs by applying V&V throughout the DT development life cycle. The framework consists of iterative, recursive steps that create a methodology for considering the elements needed to develop trusted, integrated DTs.

The DTV breaks down into 3 stages: decomposition, development, and integration and testing. In stage 1, decomposition, the problem is defined and broken down into DT conceptual design and technical requirements. The conceptual design outlines the model structure and describes how the physical system will be mirrored digitally. The technical specifications define the DT's purpose, context of use, and data requirements. Stage 2, development, begins with the analysis of alternatives, which involves creating and implementing solutions that fulfill the previously determined capability and complexity requirements. Stage 3 is integration and testing. At this stage, the DT will be deployed to the operational environment and continuously monitored. Continuous feedback loops are put in place, running a series of tests to ensure real-time feedback and recalibration. Verification ensures each iteration of the DT fulfills the design and technical requirements. There are two types of verification: model verification ensures the DT's components meet performance requirements, while code verification includes static and dynamic condition testing. Simultaneously, validation compares each iteration's performance and accuracy against the real system's performance, using comparison experiments that compare DT outputs to real system data, or thorough uncertainty quantification that measures aleatory and epistemic uncertainties. Upon the conclusion of verification and validation, the DT development is complete. The DTV requires documentation of V&V evidence to demonstrate trustworthiness, reproducibility, and transparency. The three stages of the DTV establish a replicable process that embeds V&V in every phase of DT development and provides a baseline for ongoing updates and long-term model optimization. Figure 5 shows the structure and stages of the DTV, with the V&V steps emphasized.

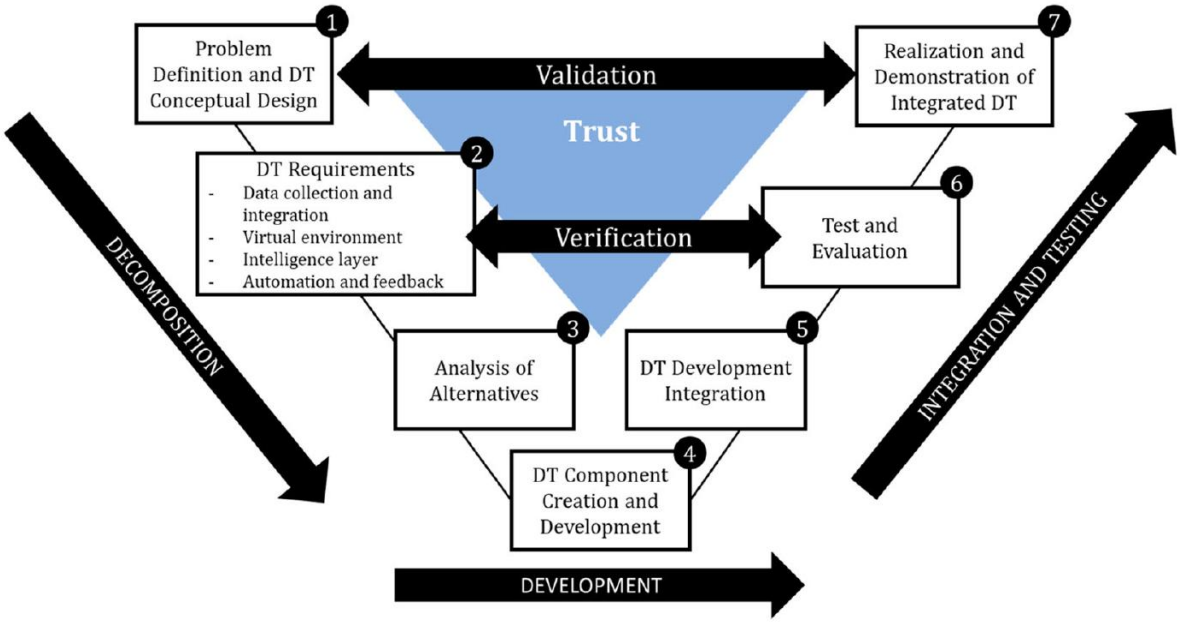


Figure 5: Digital Twin V.

6. References

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